

MOHAMATH NALEEM ZAKY AHAMED

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🌐 LinkedIn Profile | 🐙 GitHub Profile | 🌐 Portfolio

SUMMARY

Computer Science and Engineering undergraduate at the University of Moratuwa with industry experience in regulated fintech systems and strong research exposure in machine learning-driven IoT networking. Experienced in full-stack development, systems programming, and applied ML, with a proven ability to deliver end-to-end software solutions in production and research environments.

EDUCATION

University of Moratuwa

2022 – Present

B.Sc. Eng. (Hons) in Computer Science & Engineering

Royal College - Colombo 07

2012 – 2020

G.C.E. Advanced Level (2020)

3A | IR – 316

G.C.E. Ordinary Level (2017)

9A

EXPERIENCE

IronOne Technologies

Dec 2024 – Sep 2025

Software Engineering Intern

- Developed and delivered production-grade full-stack modules for the Primary Dealer System, a regulated fintech platform for government securities trading.
- Contributed to system architecture discussions, requirement analysis, and feature implementation in collaboration with business analysts and senior engineers.
- Worked across multiple development phases, supporting end-to-end feature delivery from design through implementation and testing.

Technologies: Java (Quarkus), React (TypeScript), PostgreSQL, AWS, Azure DevOps

PROJECTS

Advancing Indoor Connectivity (FYP)

Ongoing

- Developing an edge-deployed ML framework to predict LoRaWAN packet loss and improve reliability for low-power IoT nodes.
- Designing adaptive models using sequence prediction and online learning to handle dynamic indoor environments.
- Building network-level simulators incorporating urban obstacle modeling, interference analysis, supervised data generation, and reinforcement learning-based optimization.

Technologies: Python, C++ (NS3, Qt), STM32, TinyML

SalonMate – Mobile Salon Booking Platform

Ongoing

- Developing a mobile-first salon booking platform with dual applications (Customer & Owner), featuring GPS-based discovery and real-time booking.
- Implementing an offline-first architecture using SQLite/Hive and Firebase to ensure reliable data synchronization in low-connectivity environments.

Technologies: React Native, Firebase (Auth, Firestore, FCM), SQLite/Hive, Google Maps SDK

AI-Based Personalized Learning Platform

May 2025 – Jun 2025

- Engineered a tri-lingual learning platform offering adaptive lessons and personalized questions for students.
- Integrated OpenAI APIs to generate dynamic learning paths and content based on student performance.

Technologies: React.js, Node.js, MongoDB, OpenAI API

RPAL Interpreter

Apr 2024 – Jun 2024

- Built a complete interpreter for the functional programming language RPAL, consisting of a lexical analyser, parser, and a CSE machine.
- Implemented the execution pipeline by constructing the Abstract Syntax Tree (AST), standardising it, and evaluating the program to derive outputs.

Technologies: C++

Smart Security Camera Monitoring Platform

Jun 2024 – Nov 2024

- A comprehensive software solution designed to enhance surveillance capabilities through advanced image processing, featuring both web and mobile app interfaces.

Technologies: React, React Native, Node.js, Flask, PostgreSQL, YOLO

Airline Reservation System

May 2023 – Jun 2023

- Comprehensive airline reservation system that allows users to search for and book tickets, while enabling admins to generate detailed reports.

Technologies: React.js, Flask, MySQL

TECHNICAL SKILLS AND INTERESTS

Languages: C++, Python, Java, JavaScript

Frameworks: React, Quarkus, Node.js, PyTorch, OpenCV, YOLO

Databases: MySQL, MongoDB, PostgreSQL

Developer Tools: Git, GitHub, Docker, Postman

Areas of Interest: Full Stack Development, Software Engineering, Machine Learning

Spoken Languages: English (Fluent), Sinhala (Fluent), Tamil (Native)

EXTRACURRICULAR ACTIVITIES

Rotaract Club of University of Moratuwa & Alumni

2021 – 2024

Board of Directors – Rotary Family Coordinator (2022/2023).

Chairperson – General Meeting (April), Binara Padura, UN Days, Revelation (2022), and Kindred Kidneys (2023).

Served as Company Coordinator for RUR 2022 and contributed as a Graphic Designer for multiple club initiatives.

AIIESEC Colombo South

2022 – 2024

Team Leader – OGT Customer Relation Team (2022/2023); Member – IGV (2023/2024).

Coordinated with international partners, local talent, and incoming global volunteers.

Other Media & Club Activities

2023 – 2025

Mora Hiking Club: Media Committee Member (2023/2024); Graphic Designer for Stridian 2023 and general hikes. Organising Committee: StridEve 2023, Stridian (2024, 2025), Cykle (2024, 2025).

IESL Students' Chapter & CSE Media Team: Media Team Member; Graphic Designer/Video Editor for IESL AGM 2023. Designed newsletters and posters for the Computer Science & Engineering department.

Tutor at MDC (Aluthgama)

Active

Teaching Python and Web Development to school students.

REFERENCES

Prof. Dulani Meedeniya

Senior Lecturer

Department of Computer Science and Engineering

University of Moratuwa

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